

Bridging Tradition and Technology: Human-AI Interface for Exploration and Co-Creation of Classical Dance Heritage

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Abstract

This paper introduces Text2Tradition, a system designed to bridge the epistemological gap between modern language processing and traditional dance knowledge by translating user-generated prompts into Thai classical dance repertoire. Our system interprets user prompts through the lens of Mae Bot Yai—the 59 foundational movements constituting the vocabulary of traditional Thai dance—and incorporates six choreographic elements that encode centuries of cultural knowledge. This research explores the fertile tension between two knowledge systems: the embodied, culturally-specific wisdom of traditional dance and the data-driven, statistically-derived, and often Western-centric intelligence of LLMs. By mediating between these epistemologies, we highlight the potential of AI-mediated systems not only to preserve traditional forms but also to foster new cultural co-creations, suggesting that these tensions can be harnessed to stimulate cultural dialogue and innovation.

CCS Concepts

• **Applied computing** → **Arts and humanities; Media arts; Performing arts**; • **Human-centered computing** → **Human computer interaction (HCI); Systems and tools for interaction design**.

Keywords

Human-AI Interaction, Cultural Heritage, Cross-Cultural Co-Creation, AI-generated Choreography, Virtual Character

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1 Introduction

In our diverse global landscape, the intersection of differing epistemologies often creates cultural tensions [1, 10, 29]. Considering AI, LLMs promise universal translation of knowledge systems yet exhibit biases and oversimplifications, particularly regarding underrepresented non-Western traditions [17]. This raises critical questions about epistemological representation in AI and cross-cultural knowledge integration. Instead of attempting to eliminate unavoidable biases, we should focus on acknowledging and effectively managing them. This paper presents a perspective on how epistemological tensions can be harnessed to stimulate cultural co-creation, with each epistemology highlighting and augmenting others' limitations while helping raise new inquiries for us about our approach and perspective to our worldviews, leading to cultural development.

Our work Text2Tradition presents a case study of a mediated interface between two artifacts: traditional Thai choreographic knowledge and LLM. By examining the tension and interplay between the two subjects and user interaction with the interface, we explore how this kind of relationship can lead to novel co-creation and deeper reflection on the complexities of cultural translation. We aim to foreground both the potential and challenges of representing and engaging with diverse forms of embodied knowledge in computational systems and societies. This paper makes the following key contributions:

- A novel interaction paradigm for AI-mediated cultural translation that bridges traditional dance knowledge and contemporary storytelling through user-generated prompts
- Design insights and technical implementation details for creating interactive systems that preserve cultural authenticity while enabling creative reinterpretation
- A framework for evaluating and managing epistemological tensions in AI-mediated cultural systems, demonstrating

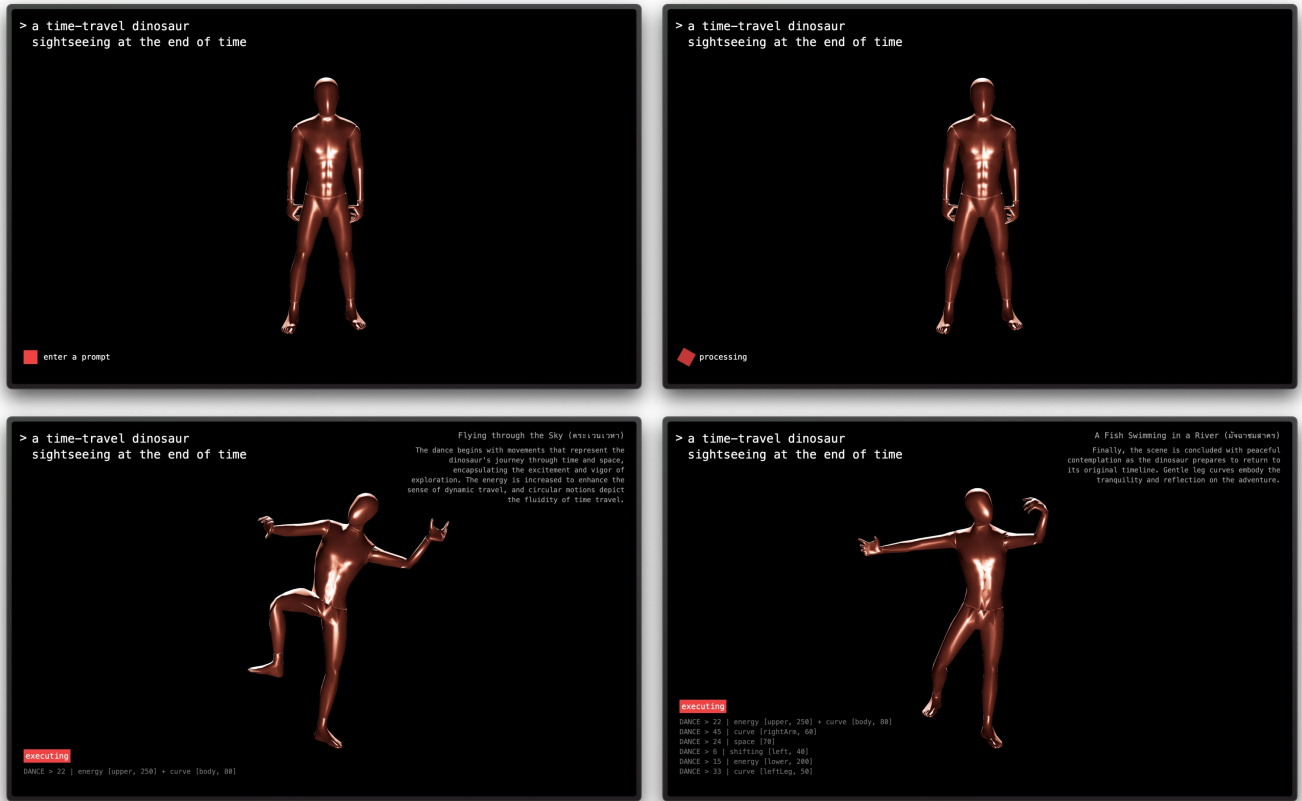


Figure 1: Text2Tradition Web Interface: A step-by-step visualization of the dance generation process, from prompt input to choreographic output. The system transforms the narrative prompt 'a time-travel dinosaur sightseeing at the end of time' into Thai classical dance interpretations, displaying both the 3D character poses and the corresponding dance interpretation.

how these tensions can foster creative innovation rather than limitation

This work advances the HCI community's understanding of how to design interactive systems that engage with cultural heritage in the age of AI. While previous HCI research has explored digital preservation of cultural artifacts and AI-generated art, our work uniquely demonstrates how thoughtful system design can enable meaningful human-AI collaboration in cultural domains. Our findings inform the design of future interactive systems that must navigate complex cultural translations, particularly in contexts where traditional knowledge systems intersect with modern technology. This research also contributes to ongoing discussions within the HCI community about responsible AI integration, participatory design with cultural communities, and the role of interactive systems in cultural preservation and evolution.

2 Background and Related Work

2.1 Dance and Computation

As a living embodiment of cultural heritage[4, 14] encoding histories and contexts of its society, the choreographic intelligence encoded in traditional dance form serves as a compelling medium

through which to explore these epistemological tensions. Prior to the consideration of cross-cultural translation, the form of dances themselves already raised important questions about the nature of creativity, embodiment, and cultural preservation in the digital age. Computational choreography has evolved significantly since Merce Cunningham's pioneering work with 3D computer graphics in the 1990s [25, 26]. The field now encompasses a wide range of approaches, including motion capture [8, 20], virtual environments [12, 13], wearable computing [15], and AI-generated dance [9, 16, 31]. These technologies have not only expanded the creative possibilities for dance composition but also opened new avenues for documenting and preserving traditional dance forms as cultural heritage [4, 14]. These approaches raise intriguing questions about the nature of creativity, the relationship between human and machine-generated choreography, and the potential for AI to bridge cultural and temporal divides in dance. However, the specific challenge of using text-based AI models to interpret and generate culturally specific dance movements remains largely unexplored.

2.2 Cultural Context

Traditional Thai dance, or Natsin, has roots in the royal courts of ancient Siam, with a history spanning centuries [24], characterized

by graceful, fluid movements, elaborate costumes, and complex hand gestures, each element carries specific meanings and symbolism [28]. Central to Thai traditional dance is "Mae Bot Yai" (The Greater Fundamentals), a collection of 59 foundational movement units. Each unit, while referred to as a pose, is actually a brief, precise movement sequence that serves as a basic alphabet in Thai classical dance composition [3, 7, 27]. A key characteristic of this art form is that each pose encodes loose meanings, with a dance repertoire essentially being a composition and morphing of these alphabets¹

However, critics have noted a gradual loss of inner interpretability and dynamism in traditional Thai dance, arguing that movements have become self-stereotyped [30]. As a result, some contend that dance has not significantly evolved since the early 20th century [6], raising questions about cultural preservation versus artistic innovation. From this, a countermovement has recently emerged [18], aiming to reinvigorate traditional Thai dance by fostering innovation within its framework. Thai choreographer Pichet Klunchun has been a significant contributor to this movement. Klunchun's project "No. 60," which serves as a foundation for our research, represents a culmination of his two-decade-long exploration of Thai classical dance [27].

Through his analysis and interpretation of Mae Bot Yai, Klunchun identified "Six Elements", the six fundamental components that form the core of traditional Thai dance movements: *Energy*, which governs the dynamic range and vitality of motion, particularly in knee movements and rhythmic stamping; *Circles & Curves*, emphasizing flowing, rounded movement pathways over linear patterns; *Axis Points*, focusing on pivotal joint positions that enable symbolic hand gestures, an elemental characteristic of traditional Thai dance; *Synchronous Limbs*, coordinating different body parts in symmetrical and asymmetrical patterns; *External Body Spaces*, considering the geometric shapes and spatial relationships created around the dancer; and *Shifting Relations*, managing smooth transitions between movements to maintain continuous flow. Klunchun then applied this system to extend the movements beyond the traditional 59 movements of Mae Bot Yai.

Building upon the No. 60, Klunchun's 2024 performance "Cyber Subin" integrates human dancers with generative virtual characters. In this regard, the open-source computational component *OpenCyberDance*² was developed as a platform that translates the Six Elements into algorithmic rules, allowing users to manipulate pre-recorded 3D virtual characters and create new digital poses [21].

3 Methodology

3.1 Text2Tradition

Text2Tradition extends *OpenCyberDance*'s for pose transformation by exploring how these generated poses can be sequenced and combined to create dance repertoires. Recognizing traditional Thai dance as a composition of movement alphabets, we explore contemporary interpretations of this art form. The platform enables users to input a story prompt (e.g., "A swan dancing"), which the

system translates into a repertoire of Mae Bot Yai movements (such as "A Swan Gliding," "Flying Through the Sky," "A Dragon Playing in the Water"). Each movement can be further refined using the six elements system, for instance, lowering lower body energy to emphasize a swan's gentle elegance. Text2Tradition is a publicly accessible web platform designed to foster broader engagement in cultural co-creation. The dance generation process is illustrated in Figure 2.

- (1) **3D Character Library:** Unlike Cyber Subin, which used 3D recordings of existing dance repertoires (e.g., Kukpat and Yok Rob), Text2Tradition utilizes Mae Bot Yai movements as a dance alphabet. We conducted 3D motion-capture sessions, specifically through Axis Studio and the Perception Neuron suit, with Thai traditional dance experts to record all 59 Mae Bot Yai movements. The sensor data, focusing on 17 primary axis points on the human body, were recorded and processed to construct a .BVH data file. The captured data was then mapped onto a virtual character by aligning the .BVH data to the virtual model's skeleton. The files were then converted to .GLB format, which were then optimized using glTF transform.
- (2) **Dance Generator:** A dance generation system utilized GPT-4o to transform user prompts into Thai classical dance sequences. The system analyzes the distinctive characteristics of each Mae Bot Yai movement while integrating the Six Elements principle as contextual framework. It outputs a structured JSON containing a choreographed sequence of Mae Bot Yai movements, each refined with Six Elements parameters that provide nuanced adjustments to align the performance with the narrative prompt.
- (3) **Web Interface:** Built on the *OpenCyberDance* technological stack, the user-facing component is a web application using Vue and Three.js. It allows users to input story prompts, view generated dance sequences, and inspect the generator's interpretation. When a user runs a prompt, the dance generator is initiated. The resulting JSON object is then rendered on the web interface, where the renderer engine and rule-based Six Element system are implemented.

4 Results & Discussion

Text2Tradition highlights the potential of collaborative, interdisciplinary approaches in AI-mediated cultural exploration. This section presents three sample prompts and results from Text2Tradition, representing different types of stories and interpretations: (1) Star Wars: A New Hope, retold as a traditional Thai Dance, (2) Lalisa dancing for a TikTok video, and (3) A steampunk adaptation of Stravinsky's The Rite of Spring. By embracing the interplay between AI systems and cultural contexts, it transforms perceived limitations into opportunities for dialogue and innovation. Considering dance generation, the process unfolds in three interconnected layers: the creative "translation" between cultures, the dynamic interaction of diverse epistemologies, and the resulting co-creation of new cultural expressions.

¹For instance, the pose "Pisamai Riang Mon" conveys meanings related to love and relationships. Choreographers select and subtly modify these poses to express specific narratives for their compositions.

²<https://github.com/mitmedialab/opencyberdance>

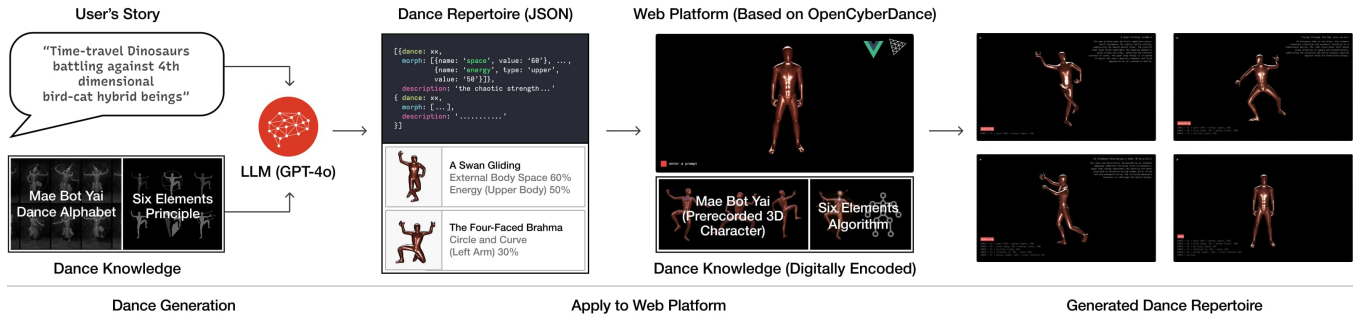


Figure 2: Dance Generation Process: The system takes a story prompt and dance knowledge as inputs to a Large Language Model (LLM), which then produces structured JSON data containing corresponding Mae Bot Yai movements, Six Element refinement parameters, and interpretations. These are subsequently visualized as a 3D character animation in the web platform.

4.1 Narratives

As depicted in Figure 3, (1) is arguably the most straightforward, where the input is a linear story translated into another form of representation. The generated data is temporally respective to the storyline, with each pose representing significant moments of the story, both imitating characters (such as Princess Leia and Luke) and events (Destruction of the Death Star). The prompt of (2) is more conceptual, lacking a definite temporal storyline. In this and similar cases, the system usually generates a dance where each pose represents a metaphor for different aspects of the concept; in (2), each pose deconstructs Lalisa’s characteristics and dance style. (3) is a blend between a linear story and a conceptual idea/theme, resulting in a new conceptual input. The result reflects this characteristic, with the generated output mixing parts of the storyline (such as "the sacrificial dance of the ritual") and conceptual representation ("ritualistic 'awakening' of the machine").

4.2 Metaphors and Representations

Text2Tradition generates dance representations that reflect multiple layers of conceptual metaphors [5]. The primary layer directly translates narrative elements into dance forms, while a secondary layer emerges from the intersection of cultural encodings and conceptual representations. As different cultures encode concepts uniquely in their artistic "alphabets," this AI-mediated space creates novel mappings of abstract metaphors [23], yielding fresh understandings. In the Star Wars example, the Death Star’s destruction is depicted as an elephant destroying a shed. This metaphor may embody the Death Star’s scale, represent the Rebel forces’ strength, and mirror the explosion’s spectacle within traditional contexts. This rich metaphor imbues the event with cultural significance while conveying the idea of a powerful force overcoming an impenetrable structure. Text2Tradition thus bridges disparate cultural and conceptual frameworks, offering new perspectives on familiar narratives through layered, culturally informed physical metaphors.

4.3 Lost in Translation?

While information loss is inherent in any translation process, including our system with its 59 Mae Bot Yai movements and six elements, viewing this solely as "lost in translation" may overlook

the creative possibilities that emerge through this cultural interchange. It may be more constructive to consider translation not as unidirectional. That is, in an ecosystem of such a kind, the original input is not lost but complemented by the translation process. The resulting dance sequence should be seen not as an imperfect translation but as a new creation inspired by and in dialogue with the original story. This shifts our focus from "what might be lost?" to "what is gained?" through creative interpretation.

Reframing the discussion, the cultural context, and substitutions in the translated product highlight the system’s constraints. However, these limitations may foster novel interpretations and unexpected connections formed by the interplay between the subjects in the translation process. In the case of Text2Tradition, for example, the user’s story, the LLM translator, and the traditional dance form.

4.4 Interpretation and Epistemological Interplay

The dance generation process involves three key aspects of interpretation: (1) creating new stories that transcend traditional contexts, (2) selecting Mae Bot Yai movements based on their encoded meanings within the repertoire, and (3) morphing individual movements using the rule-based six elements to emphasize their significance relative to the narrative. This means that the translation process is not a direct, one-fold translation but a multistage process, with each encoding its contextual factors. The interaction between these processes leads to a crossing of cultural contextual information, where each step loses and gains information, resulting in a dynamic system that yields new meanings and creative outputs.

Users also play a crucial role in this design, not merely as story creators and observers but as active interpreters themselves. They should be encouraged to examine and rethink the implications of the translated product, understanding it not as an absolute answer but as a signal for further examination, interpretation, and reconsideration.

4.5 Toward Cross-Cultural Co-Creation

Cultural stagnation has become an increasing concern in various domains [2, 4, 22], including traditional Thai dance [11], where the art form risks losing relevance in contemporary society. Text2Tradition,

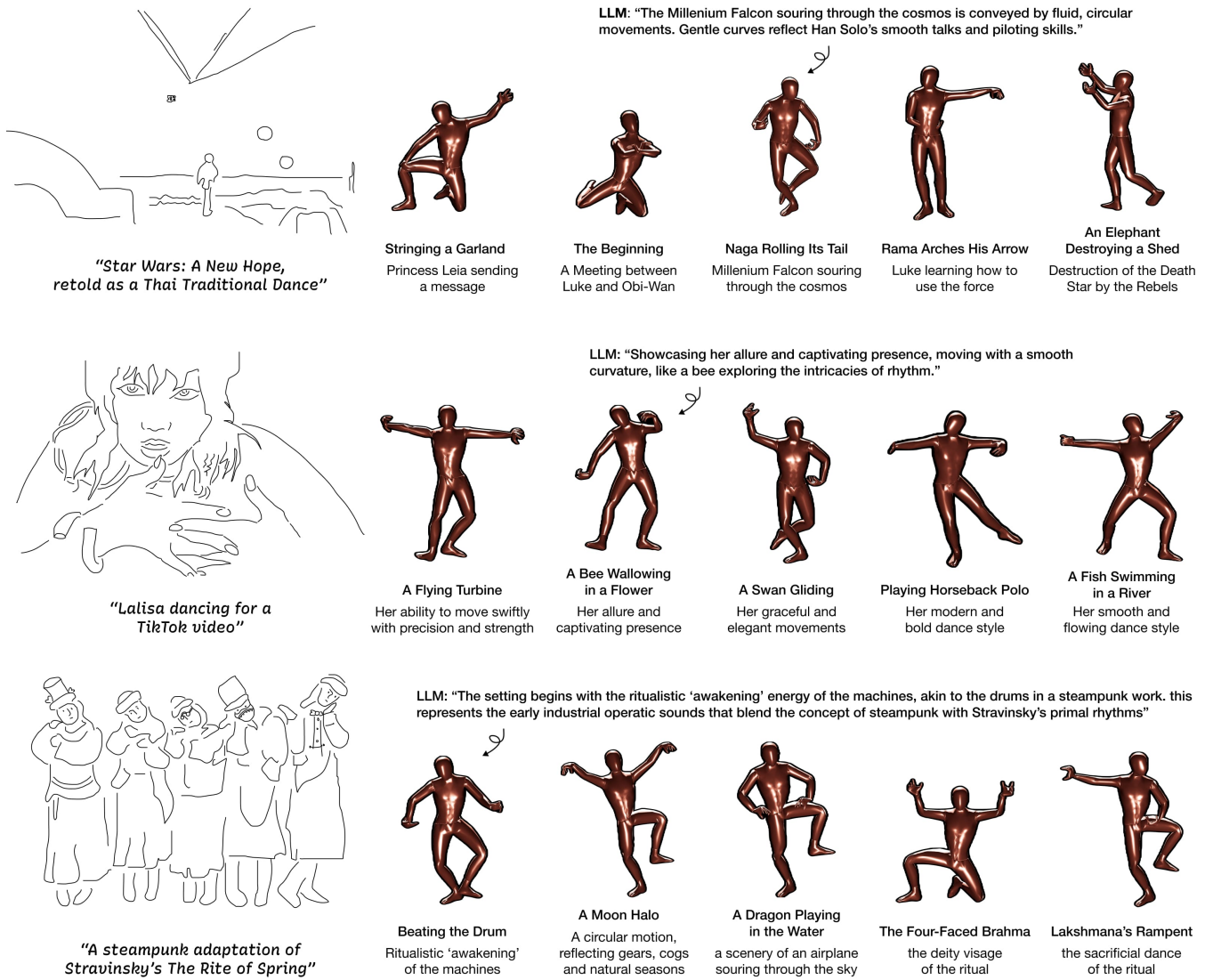


Figure 3: Three Generated Dances from Three Contemporary Tales — Star Wars: A New Hope, Lalisa's TikTok performance, and a steampunk adaptation of Stravinsky —each visualized as a series of Mae Bot Yai movement with corresponding movement descriptions that translate modern stories into traditional Thai dance repertoire.

like the work of Klunchun and other experimental preservationists [19], aims to encourage engagement that fosters new perspectives and deepens cultural dialogue, allowing different epistemologies to learn from and evolve with each other. In *Cyber Subin* and *Text2Tradition*, rather than providing a static product, the systems offer dynamic tools for interpretation that can catalyze new co-creations between users and AI systems. The technological approach has a crucial effect: the democratization of cultural interpretability, which reduces the friction to cultural knowledge and enables broader societal participation in a multi-layered interpretative process, contributing to the ongoing evolution of both traditional dance forms and contemporary narrative expression.

The result can be viewed as a technologically-induced cultural synthesis, blending different cultural artifacts and epistemologies and bridging the gap between preserved traditions and modern matters.

4.6 Future Directions

4.6.1 Expanding the Cultural Scope & Enhancing Interactivity and Embodiment. While our project focused specifically on Thai classical dance, the methodologies developed here could potentially be applied to other traditional art forms facing similar challenges of preservation and contemporary relevance. Future research could explore the application of this approach to diverse cultural traditions, contributing to a more globally inclusive development of AI in the arts. In addition, future iterations of the *Text2Tradition* system

could explore more interactive and embodied forms of engagement. This might include the integration of motion capture technology to allow real-time interaction between human dancers and the AI system or the development of augmented reality applications that could overlay AI-generated movements onto physical spaces.

4.6.2 Considerations and Ethical Frameworks for AI in Cultural Domains. AI systems like Text2Tradition enable cross-cultural co-creation but demand ongoing reevaluation to mitigate cultural oversimplification from Western-centric training data. While democratizing cultural interpretation is valuable, caution is necessary to prevent imposing dominant narratives or risking algorithmic colonization [1]. Ethical frameworks for AI in cultural domains should emphasize inclusive design, diverse perspectives, and ownership rights, ensuring source communities maintain control over their cultural expressions. Future work should prioritize participatory approaches, involving communities in the co-creation process to enhance cultural authenticity and relevance.

5 Conclusion

Through Text2Tradition, this paper demonstrates how interactive systems can enable meaningful human-AI collaboration in cultural domains. The system transforms Thai classical dance into contemporary expressions while preserving cultural authenticity, illustrating technology's dual role in cultural preservation and evolution. This approach offers a framework for managing epistemological tensions in AI-mediated cultural systems, balancing preservation with creative reinterpretation.

The findings contribute to HCI research by demonstrating effective approaches to cultural translation through technology. This work provides valuable insights for designing future interactive systems in complex cultural domains, advancing discourse on responsible AI integration and participatory design with cultural communities.

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